Student Name: Student ID:

# ELEC 473 – Cryptography and Network Security
# Quiz 2

Wednesday, Oct. 25, 2023, 2:30-3:00 PM

INSTRUCTIONS—Read Carefully

• Write your name and ID legibly at the top of the papers.

• Calculators are allowed to use.

• Show all your work. Partial credit will be given for substantial progress towards the solution. No credit will be given for answers with no substantial process.

**Part I Multiple Choices - please mark the best answer (subtotal: 10 marks).**

1. Data encryption standard (DES) is a block cipher and encrypts data in blocks of size of _______?

B

A. 56
B. 64
C. 128
D. 32

C

2. Which one of the following statements is false about the block cipher?
A. Block ciphers can be used as the keystream generators in stream cipher.
B. A plaintext block is used to produce a ciphertext block.
C. Confusion hides the relationship between the ciphertext and the plaintext.
D. If we change a single bit of the plaintext, then (statistically) half of the bits in the ciphertext should change.

C

3. Give the following DES S-box, if the input is 100100, what is the output of the S-box?
A. 1110
B. 1101
C. 1000
D. 1001

| | 0000 | 0001 | 0010 | 0011 | 0100 | 0101 | 0110 | 0111 | 1000 | 1001 | 1010 | 1011 | 1100 | 1101 | 1110 | 1111 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 00 | 0000 | 0001 | 0010 | 0011 | 0100 | 0101 | 0110 | 0111 | 1000 | 1001 | 1010 | 1011 | 1100 | 1101 | 1110 | 1111 |
| 01 | 0101 | 0100 | 0111 | 0110 | 0001 | 0000 | 0011 | 0010 | 1101 | 1100 | 1111 | 1110 | 1001 | 1000 | 1011 | 1010 |
| 10 | 1010 | 1011 | 1000 | 1001 | 1110 | 1111 | 1100 | 1101 | 0010 | 0011 | 0000 | 0001 | 0110 | 0111 | 0100 | 0101 |
| 11 | 1111 | 1110 | 1101 | 1100 | 1011 | 1010 | 1001 | 1000 | 0111 | 0110 | 0101 | 0100 | 0011 | 0010 | 0001 | 0000 |

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D 4. Which of the following is an encryption mode for the Block ciphers in cryptography?
A. Electronic Code Book (ECB)
B. Counter (CTR) mode
C. Cipher Block Chaining (CBC)
D. All of the above

C 5. Which of the following is the main disadvantage of the ECB (Electronic Code Book)?
A. It requires large block size.
B. Padding is done to make the plaintext divisible into blocks of fixed size.
C. It is prone to cryptanalysis since there is a direct relationship between plaintext and ciphertext.
D. None of the above.

A 6. Which of the following is the structure of DES?
A. Feistel network
B. Luby rackoff block cipher
C. Asymmetric cipher
D. substitution-permutation network

D 7. Which of the following is the key size of AES?
A. 128
B. 256
C. 192
D. All of the above

A 8. 128 bits key form of AES has __ rounds?
A. 10
B. 12
C. 14
D. All of the above

C 9. Which of the following is not a group?
A. (Z, +)
B. ($Z_n$, +)
C. ($Z_n$, x)
D. ($Z_n$*, x)

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B

10. What is the order of the multiplicative group ($Z_{35}$*, x)?
A. 34.
B. 24.
C. 35.
D. 28.

**Part II (subtotal: 20 marks).**
Q1.(8 marks) Solving the following system of congruences:

$$3x \equiv 1 \bmod 7$$

$$9x \equiv 5 \bmod 11$$

$$7x \equiv 5 \bmod 13$$

gcd(7, 11) = 1, gcd(7, 13) = 1, gcd(11, 13) = 1. (1')

$3^{-1} \bmod 7 = 5$. $9^{-1} \bmod 11 = 5$. $7^{-1} \bmod 13 = 2$.

$$\begin{cases} 3x \equiv 1 \bmod 7 \\ 9x \equiv 5 \bmod 11 \\ 7x \equiv 5 \bmod 13 \end{cases} \Rightarrow \begin{cases} 3^{-1} \cdot 3x \equiv 3^{-1} \cdot 1 \bmod 7 \\ 9^{-1} \cdot 9x \equiv 9^{-1} \cdot 5 \bmod 11 \\ 7^{-1} \cdot 7x \equiv 7^{-1} \cdot 5 \bmod 13 \end{cases} \Rightarrow \begin{cases} x \equiv 5 \bmod 7 & (1') \\ x \equiv 3 \bmod 11 & (1') \\ x \equiv 10 \bmod 13 & (1') \end{cases}$$

$M = 7 \times 11 \times 13 = 1001$

$M_1 = 143$. $M_2 = \cancel{77}\,91$. $M_3 = 77$

$y_1 = 143^{-1} \bmod 7 = 5$ (1') $y_2 = 91^{-1} \bmod 11 = 4$ (1') $y_3 = 77^{-1} \bmod 13 = 12$ (1')

$x = 5 \times 143 \times 5 + 4 \times 91 \times 3 + 12 \times 77 \times 10 \bmod 1001$

$= 894 \bmod 1001$. (1')

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Q2. (8 marks) a) Please give all the elements in the multiplicative group ($Z_{14}$*, x)?

b) Please give the order of each element a ∈ $Z_{14}$*?

c) What is Euler's totient function of n=198?

a) $Z_{14}$*={1, 3, 5, 9, 11, 13} (3')

b)

| $Z_{14}$* | 1 | 3 | 5 | 9 | 11 | 13 |
|---|---|---|---|---|---|---|
| Ord(a) | 1 | 6 | 6 | 3 | 3 | 2 |

(3')

Hint: ord(a)|Ord($Z_{14}$*), Ord($Z_{14}$*)=6.

c. $\varphi(p_1^{a_1}p_2^{a_2}\cdots p_t^{a_t}) = p_1^{a_1}(1-\frac{1}{p_1})p_2^{a_2}(1-\frac{1}{p_2})\cdots p_t^{a_t}(1-\frac{1}{p_t})$

φ(n)= φ(198)= φ($2*3^2*11$)=60

(2')

Q3: (4 marks) Use the **Repeated Squaring Algorithm** to compute $5^{20}$ in $Z_{23}$*.

20=$(10100)_2$ (1')

| | Y | Z |
|---|---|---|
| | 5 | 1 |
| x0=0 | $5^2$ mod 23=2 | 1 |
| X1=0 | $2^2$mod 23=4 | 1 |
| X2=1 | $4^2$mod 23=16 | 4*1mod23=4 |
| X3=0 | $16^2$mod23=3 | 4 |
| X4=1 | $3^2$mod23=9 | 3*4 mod 23=12 |

(2')

Thus, $5^{20}$ mod 23=12

(1')